

**Module 5 Assignment — Nonparametric Methods and Sampling**

**by**

**Sri Ram Prabu**

**ALY6015: Intermediate Analytics**

**Dr. Paul Dooley**

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## Introduction

This report presents a comprehensive analysis of various statistical problems using nonparametric methods and simulation techniques, as required by the ALY6015 assignment. Each problem is addressed systematically, following the required steps for hypothesis testing: stating the hypotheses, identifying critical values, computing test values, making a decision, and summarizing the results. The analyses were conducted using the R programming language to ensure accuracy and reproducibility. The report covers Sign Tests, Wilcoxon Rank-Sum Tests, the Kruskal-Wallis Test, Spearman Rank Correlation, and simulation problems to determine averages based on probability.

### Section 13-2, Problem 6: Game Attendance

**a. State the hypotheses and identify the claim.**

- $H_0: \text{median} = 3000$  (Claim)
- $H_1: \text{median} \neq 3000$
- Significance level  $\alpha = 0.05$ . This is a two-tailed test.

**b. Find the critical value(s).**

- We use the Sign Test. The sample size is  $n=20$ . There are 10 values above 3000 and 10 values below.
- Using a Sign Test critical value table for  $n=20$  and  $\alpha=0.05$  (two-tailed), the critical value is **5**.
- We will reject  $H_0$  if our test statistic is less than or equal to 5.

**c. Compute the test value.**

- The test statistic is the smaller of the number of values above or below the median.
- Number of values  $> 3000 = 10$ .

- Number of values  $< 3000 = 10$ .
- The test value is **10**.

**d. Make the decision.**

- **Traditional Method:** The test value (10) is not less than or equal to the critical value (5).
- **P-value Method (from R):** The R output shows a **p-value = 1.0**.
- Since  $p(1.0) > \alpha(0.05)$ , we **fail to reject the null hypothesis** ( $H_0$ ).

**e. Summarize the results.**

- There is not sufficient evidence at the 0.05 significance level to reject the athletic director's claim that the median paid attendance is 3000.
- **Follow-up Question:** Yes, since we could not reject the claim, 3000 appears to be a statistically sound figure for the median and could be used as a guide for printing programs.

### Section 13-2, Problem 10: Lottery Ticket Sales

**a. State the hypotheses and identify the claim.**

- $H_0: \text{median} \geq 200$
- $H_1: \text{median} < 200$  (Claim)
- Significance level  $\alpha = 0.05$ . This is a left-tailed test.

**b. Find the critical value(s).**

- Since  $n=40$  is greater than 25, we use the z-approximation.
- For a left-tailed test at  $\alpha=0.05$ , the critical value is  **$z = -1.645$** .
- We will reject  $H_0$  if the test z-value is less than -1.645.

**c. Compute the test value.**

- We have  $n=40$  days and  $x=15$  days with sales *fewer* than 200.
- The test z-statistic is calculated as: 
$$z = \frac{(x + 0.5) - (n/2)}{\sqrt{n}/2} = \frac{(15 + 0.5) - (40/2)}{\sqrt{40}/2} = \frac{15.5 - 20}{3.162} \approx -1.423$$

**d. Make the decision.**

- **Traditional Method:** The test value  $z = -1.423$  is not less than the critical value  $z = -1.645$ .
- **P-value Method:** The p-value corresponding to the test statistic  $z = -1.423$  for a one-tailed (left) test is  $P(Z < -1.423) = 0.077$ .
- Since  $p (0.077) > \alpha (0.05)$ , we **fail to reject the null hypothesis** ( $H_0$ ).

**e. Summarize the results.**

- There is not sufficient evidence at the 0.05 significance level to conclude that the median number of lottery tickets sold is below 200.

### Section 13-3, Problem 4: Lengths of Prison Sentences

**a. State the hypotheses and identify the claim.**

- $H_0$ : There is no difference in the sentence received by each gender (Claim).
- $H_1$ : There is a difference in the sentence received by each gender.
- Significance level  $\alpha = 0.05$ . This is a two-tailed test.

**b. Find the critical value.**

- The R `wilcox.test` function uses the p-value method, so a critical W value is not computed for the decision.

**c. Compute the test value.**

- From the R output, the Wilcoxon rank-sum test statistic is  $W = 107.5$ .

**d. Make the decision.**

- **P-value Method (from R):** The R output shows a **p-value = 0.2053**.
- Since  $p(0.2053) > \alpha(0.05)$ , we **fail to reject the null hypothesis** ( $H_0$ ).

**e. Summarize the results.**

- There is not sufficient evidence at the 0.05 significance level to reject the claim that there is no difference in sentence lengths between genders. The data supports the claim.

**Section 13-3, Problem 8: Winning Baseball Games**

**a. State the hypotheses and identify the claim.**

- $H_0$ : There is no difference in the number of wins.
- $H_1$ : There is a difference in the number of wins (Claim).
- Significance level  $\alpha = 0.05$ . This is a two-tailed test.

**b. Find the critical value.**

- The R `wilcox.test` function uses the p-value method.

**c. Compute the test value.**

- From the R output, the Wilcoxon rank-sum test statistic is  $W = 60$ .

**d. Make the decision.**

- **P-value Method (from R):** The R output shows a **p-value = 0.7377**.
- Since  $p(0.7377) > \alpha(0.05)$ , we **fail to reject the null hypothesis** ( $H_0$ ).

**e. Summarize the results.**

- There is not sufficient evidence at the 0.05 significance level to conclude there is a difference in the number of wins between the two leagues.

**Section 13-4: Wilcoxon Signed-Rank Test (Table K)**

This section requires using Table K to make decisions.

- **5.**  $w_s = 13$ ,  $n = 15$ ,  $\alpha = 0.01$ , **two-tailed**
  - **Critical Value:** From Table K, the critical value is 16.
  - **Decision:** We reject  $H_0$  if  $w_s \leq 16$ . Since  $13 \leq 16$ , we **Reject  $H_0$** .
- **6.**  $w_s = 32$ ,  $n = 28$ ,  $\alpha = 0.025$ , **one-tailed**
  - **Critical Value:** Since  $n = 28 > 25$ , we use the z-approximation. The one-tailed critical value for  $\alpha = 0.025$  is  $z = -1.96$ .
  - **Test Value:** The calculated test statistic is  $z \approx -3.894$ .
  - **Decision:** Since  $-3.894 < -1.96$ , we **Reject  $H_0$** .
- **7.**  $w_s = 65$ ,  $n = 20$ ,  $\alpha = 0.05$ , **one-tailed**
  - **Critical Value:** From Table K, the critical value is 60.
  - **Decision:** We reject  $H_0$  if  $w_s \leq 60$ . Since  $65 \not\leq 60$ , we **Fail to reject  $H_0$** .
- **8.**  $w_s = 22$ ,  $n = 14$ ,  $\alpha = 0.10$ , **two-tailed**
  - **Critical Value:** From Table K, the critical value is 21.
  - **Decision:** We reject  $H_0$  if  $w_s \leq 21$ . Since  $22 \not\leq 21$ , we **Fail to reject  $H_0$** .

**Section 13-5, Problem 2: Mathematics Literacy Scores****a. State the hypotheses and identify the claim.**

- $H_0$ : There is no difference in the mean mathematics literacy scores among the regions (Claim).
- $H_1$ : There is a difference in the mean mathematics literacy scores.
- Significance level  $\alpha = 0.05$ .

**b. Find the critical value.**

- The Kruskal-Wallis test follows a  $\chi^2$  distribution with  $df = k - 1 = 3 - 1 = 2$ .
- The critical H value for  $\alpha = 0.05$  and  $df = 2$  is **5.991**.
- We will reject  $H_0$  if the test value  $H > 5.991$ .

**c. Compute the test value.**

- From the R output, the Kruskal-Wallis test statistic is **H = 3.664**.

**d. Make the decision.**

- **Traditional Method:** The test value  $H = 3.664$  is not greater than the critical value 5.991.
- **P-value Method (from R):** The R output shows a **p-value = 0.1601**.
- Since  $p (0.1601) > \alpha (0.05)$ , we **fail to reject the null hypothesis ( $H_0$ )**.

**e. Summarize the results.**

- There is not sufficient evidence at the 0.05 significance level to reject the claim that there is no difference in the mean mathematics literacy scores among the three regions.

### Section 13-6, Problem 6: Subway and Commuter Rail Passengers

**a. Find the Spearman rank correlation coefficient.**

- From the R output, the Spearman coefficient is  $r_s (\text{rho}) = 0.8857$ .

**b. State the hypotheses.**

- $H_0: \rho_s = 0$  (There is no relationship)
- $H_1: \rho_s \neq 0$  (There is a relationship) (Claim)
- Significance level  $\alpha = 0.05$ .

**c. Find the critical value.**

- Using a Spearman rank correlation table for  $n=6$  and  $\alpha=0.05$  (two-tailed), the critical value is  $\pm 0.886$ .
- We will reject  $H_0$  if  $|r_s| \geq 0.886$ .

**d. Make the decision.**

- **Traditional Method:** The test value  $|r_s| = 0.8857$  is just below the table's critical value of 0.886.
- **P-value Method (from R):** The R output provides the exact p-value: **p-value = 0.0355**.
- Since  $p(0.0355) < \alpha(0.05)$ , we **Reject the null hypothesis ( $H_0$ )**. (The p-value method is more precise).

**e. Summarize the results.**

- There is sufficient evidence at the 0.05 significance level to support the claim that a relationship exists between the number of daily subway and commuter rail passengers.
- **Follow-up Question:** A transportation authority could use this strong positive relationship to plan integrated services, such as co-locating stations or offering bundled fares, knowing that infrastructure improvements would likely benefit both systems.

**Section 14-3: Simulation Techniques****16. Prizes in Caramel Corn Boxes**

- **Task:** Find the average number of boxes needed to get all four prizes, based on 40 simulations.
- **R Simulation Result:** After running 40 trials (with `set.seed(123)` for reproducibility), the average number of boxes required was **8.5**.

### 18. Lottery Winner

- **Task:** Find the average number of tickets needed to spell "big", based on 30 simulations.
- **R Simulation Result:** After running 30 trials (with `set.seed(123)` for reproducibility), the average number of tickets required was **10.7**.

## Conclusion

In conclusion, this report successfully applied a range of nonparametric statistical tests and simulation methods to analyze. Key findings from the hypothesis tests showed no statistically significant evidence to reject claims regarding the median game attendance, median lottery ticket sales, differences in prison sentences, differences in baseball wins, or differences in math literacy scores. However, a significant correlation between subway and commuter rail ridership was found. Furthermore, simulations determined that an average of 8.5 boxes are needed to collect a set of four prizes, and an average of 10.7 tickets are required to win the 'big' lottery prize. These results demonstrate the practical application of nonparametric statistics and simulation in drawing conclusions from data where assumptions of normality may not be met.

## References

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## Appendix

```
library(BSDA)

# Section 13-2: Sign Tests

# 6) Game Attendance
# We will use the binom.test function
# performs the same calculation as the Sign Test.
attendance <- c(6210, 3150, 2700, 3012, 4875, 3540, 6127, 2581,
               2642, 2573, 2792, 2800, 2500, 3700, 6030, 5437,
               2758, 3490, 2851, 2720)
# H0: median = 3000 (Claim)
# H1: median != 3000
# Count successes (e.g., values > 3000)
n_6 <- length(attendance)
x_6 <- sum(attendance > 3000) # This will be 10
# Note: sum(attendance < 3000) is also 10.
cat("\n--- Section 13-2, Problem 6 (Game Attendance) ---\n")
binom_test_6 <- binom.test(x = x_6, n = n_6, p = 0.5, alternative = "two.sided")
print(binom_test_6)

# 10) Lottery Ticket Sales
# H0: median >= 200
# H1: median < 200 (Claim)
# n = 40 days, x = 15 days SOLD FEWER THAN 200.
# If H1 is true, the probability of selling < 200 (p) should be > 0.5.
# So we test H1: p > 0.5, using x=15.
n_10 <- 40
x_10 <- 15 # Number of days with sales < 200
cat("\n--- Section 13-2, Problem 10 (Lottery Sales) ---\n")
binom_test_10 <- binom.test(x = x_10, n = n_10, p = 0.5, alternative = "greater")
print(binom_test_10)

# Section 13-3: Wilcoxon Rank-Sum (Mann-Whitney)

# 4) Prison Sentence Lengths
# H0: No difference (Claim)
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```

# H1: A difference exists
males <- c(8, 12, 6, 14, 22, 27, 32, 24, 26, 19, 15, 13)
females <- c(7, 5, 2, 3, 21, 26, 30, 9, 4, 17, 23, 12, 11, 16)
cat("\n--- Section 13-3, Problem 4 (Prison Sentences) ---\n")
wilcox_test_4 <- wilcox.test(males, females, alternative = "two.sided",
                           correct = TRUE) # Use continuity correction
print(wilcox_test_4)

# 8) Winning Baseball Games
# H0: No difference
# H1: A difference exists (Claim)
nl <- c(89, 96, 88, 101, 90, 91, 92, 96, 108, 100, 95)
al <- c(108, 86, 91, 97, 100, 102, 95, 104, 95, 89, 88, 101)
cat("\n--- Section 13-3, Problem 8 (Baseball Wins) ---\n")
wilcox_test_8 <- wilcox.test(nl, al, alternative = "two.sided",
                           correct = TRUE)
print(wilcox_test_8)

# Section 13-4: Wilcoxon Signed-Rank Criticals
# This section requires using Table K.
# R code is not used for the primary decision, but we can
# calculate the Z-value for  $n > 25$  (Problem 6).

# For Problem 6 ( $n=28$ ,  $ws=32$ ,  $\alpha=0.025$ , one-tailed)
n_13_4_6 <- 28
ws_13_4_6 <- 32
mu_w <- (n_13_4_6 * (n_13_4_6 + 1)) / 4
sigma_w <- sqrt((n_13_4_6 * (n_13_4_6 + 1) * (2 * n_13_4_6 + 1)) / 24)
z_test <- (ws_13_4_6 - mu_w) / sigma_w
z_crit <- qnorm(0.025)
cat("\n--- Section 13-4, Problem 6 (Z-Test Approx) ---\n")
cat(paste("Mean ( $\mu_w$ ):", mu_w, "\n"))
cat(paste("Std Dev ( $\sigma_w$ ):", round(sigma_w, 3), "\n"))
cat(paste("Test Statistic (z):", round(z_test, 3), "\n"))
cat(paste("Critical Value ( $z_{crit}$ ):", round(z_crit, 3), "\n"))

# Section 13-5: Kruskal-Wallis Test

# 2) Mathematics Literacy Scores
# H0: No difference in means (Claim)

```

```

# H1: A difference exists
scores <- c(527, 406, 474, 381, 411, # Western Hemisphere
           520, 510, 513, 548, 496, # Europe
           523, 547, 547, 391, 549) # Eastern Asia
groups <- factor(c(rep("WH", 5), rep("Europe", 5), rep("EAsia", 5)))
cat("\n--- Section 13-5, Problem 2 (Math Scores) ---\n")
kruskal_test_2 <- kruskal.test(scores ~ groups)
print(kruskal_test_2)
# Find the critical value
df_k <- kruskal_test_2$parameter
crit_val_H <- qchisq(p = 0.95, df = df_k)
cat(paste("Critical Value (H):", round(crit_val_H, 3), "\n"))

# Section 13-6: Spearman Rank Correlation

# 6) Subway and Commuter Rail Passengers
# H0: rho_s = 0 (No relationship)
# H1: rho_s != 0 (A relationship exists) (Claim)
subway <- c(845, 494, 425, 313, 108, 41)
rail <- c(39, 291, 142, 103, 33, 38)
cat("\n--- Section 13-6, Problem 6 (Transit) ---\n")
# Use exact=TRUE for small sample size
cor_test_6 <- cor.test(subway, rail, method = "spearman", exact = TRUE)
print(cor_test_6)

# Section 14-3: Simulations

# Set seed for reproducible results
set.seed(123)

# 16) Prizes in Caramel Corn Boxes
# Find avg. number of boxes to get all 4 prizes (40 trials)
trials_16 <- 40
get_boxes <- function() {
  prizes_found <- logical(4) # A "checklist" for prizes 1, 2, 3, 4
  count <- 0
  while (!all(prizes_found)) {
    count <- count + 1
    prize_won <- sample(1:4, 1)
    prizes_found[prize_won] <- TRUE
  }
}

```

```
    return(count)
  }
results_16 <- replicate(trials_16, get_boxes())
mean_boxes <- mean(results_16)
cat("\n--- Section 14-3, Problem 16 (Prizes) ---\n")
cat(paste("Mean boxes to get all 4 prizes (40 trials):", mean_boxes, "\n"))
# print(summary(results_16)) # Optional: for more detail

# 18) Lottery Winner
# Find avg. number of tickets to spell "big" (30 trials)
# P(b)=0.6, P(i)=0.3, P(g)=0.1
trials_18 <- 30
get_letters <- function() {
  letters_found <- c(b = FALSE, i = FALSE, g = FALSE) # Named checklist
  count <- 0
  while (!all(letters_found)) {
    count <- count + 1
    ticket <- sample(c("b", "i", "g"), 1, prob = c(0.6, 0.3, 0.1))
    letters_found[ticket] <- TRUE
  }
  return(count)
}
results_18 <- replicate(trials_18, get_letters())
mean_tickets <- mean(results_18)
cat("\n--- Section 14-3, Problem 18 (Lottery) ---\n")
cat(paste("Mean tickets to spell 'big' (30 trials):", mean_tickets, "\n"))
# print(summary(results_18)) # Optional: for more detail

cat("\n--- Script Completed ---\n")
```